Data Science (PG) Project Report

Credit Card Default Prediction

Submitted by:

Team Name: Null Pointers

Dataset: Default of Credit Card Clients

Source: UCI Machine Learning Repository

Language: Python 3

Contents

1 Dataset Description **3**

1.1 About the Dataset . . . 3

1.2 Feature Description . . . 3

2 Data Pre-Processing **5**

2.1 Why Pre-Processing is Needed . . . 5

2.2 Steps Performed . . . 5

2.2.1 Dropping the ID Column . . . 5

2.2.2 Renaming the Target Column . . . 5

2.2.3 Fixing the EDUCATION Column . . . 5

2.2.4 Fixing the MARRIAGE Column . . . 5

2.2.5 Remapping the SEX Column . . . 5

2.2.6 One-Hot Encoding . . . 6

2.2.7 Handling Infinity Values . . . 6

3 Descriptive and Statistical Analysis **7**

3.1 Basic Summary Statistics . . . 7

3.2 Default Rate Across Groups . . . 7

3.3 Correlation with Default . . . 8

4 Visualisations and Inferences **9**

4.1 Target Class Distribution . . . 9

4.2 Default Rate by Education and Gender . . . 9

4.3 Age Distribution . . . 10

4.4 Credit Limit by Default Status . . . 10

4.5 Repayment Status Analysis . . . 11

4.6 Correlation Heatmap . . . 12

4.7 Bill Amount and Payment Trends Over 6 Months . . . 13

5 Hypothesis and Research Questions **14**

5.1 Hypotheses Framed . . . 14

5.2 Research Questions . . . 14

6 Feature Engineering **15**
6.1 Why We Created New Features 15
6.2 Engineered Features 15
6.3 Feature Distributions 16

7 Machine Learning Classification **17**
7.1 Algorithm Selection and Justification 17
7.2 How the Models Were Trained 17

8 Model Evaluation and Results **18**
8.1 Performance Comparison 18
8.2 ROC Curves 19
8.3 Confusion Matrices 19
8.4 XGBoost Classification Report 20
8.5 Cross-Validation 20

9 Feature Importance **21**
9.1 Key Observations 21

10 Conclusions **23**
10.1 Summary of Findings 23
10.2 Hypotheses Revisited 24
10.3 Research Questions Answered 24
10.4 Business Recommendation 24

Appendix: Code Summary **26**

Chapter 1. Dataset Description

1.1 About the Dataset

The dataset used in this project is the **Default of Credit Card Clients** dataset, originally published by I-Cheng Yeh. It is publicly available on the UCI Machine Learning Repository and also on Kaggle.

The dataset contains records of 30,000 credit card customers from a bank in Taiwan. The data covers the period from April 2005 to September 2005. The main goal is to predict whether a customer will default on their credit card payment in the following month.

Property	**Value**
Total Rows	30,000
Total Columns	25
File Format	XLS
Target Variable	default payment next month (0 or 1)
Overall Default Rate	22.1%

1.2 Feature Description

The dataset has 25 columns in total. Below is a description of each feature:

Feature	**Description**
ID	Unique identifier for each customer
LIMIT_BAL	Credit limit given to the customer (in NT dollars)
SEX	Gender (1 = Male, 2 = Female)
EDUCATION	Education level (1 = Graduate school, 2 = University, 3 = High school, 4 = Others)
MARRIAGE	Marital status (1 = Married, 2 = Single, 3 = Others)
AGE	Age of the customer in years
PAY_0	Repayment status in September 2005
PAY_2 to PAY_6	Repayment status from August to April 2005
BILL_AMT1 to BILL_AMT6	Bill statement amounts from September to April 2005 (NT dollars)
PAY_AMT1 to PAY_AMT6	Amount paid in each month from September to April 2005 (NT dollars)
default payment next month	Target variable (1 = Default, 0 = No default)

The repayment status columns (PAY_0 to PAY_6) use the following coding: -2 means no consumption, -1 means the customer paid on time, 0 means the minimum amount was paid, and values 1 through 8 represent the number of months the payment was delayed.

Chapter 2. Data Pre-Processing

2.1 Why Pre-Processing is Needed

Raw datasets almost always have issues that can affect how well a machine learning model learns. Before building any model, we need to clean and prepare the data. In this dataset, we found several things that needed to be fixed.

2.2 Steps Performed

2.2.1 Dropping the ID Column

The ID column is just a serial number to identify rows. It carries no information about a customer's financial behaviour, so it was removed before analysis.

2.2.2 Renaming the Target Column

The target column was originally named *default payment next month*, which is long and inconvenient to use in code. It was renamed to **DEFAULT** for simplicity.

2.2.3 Fixing the EDUCATION Column

According to the dataset documentation, the valid values for EDUCATION are 1, 2, 3, and 4. However, the data also contained values 0, 5, and 6, which are not documented anywhere. These were grouped into category 4 (Others) since their actual meaning is unknown.

2.2.4 Fixing the MARRIAGE Column

Similarly, the valid values for MARRIAGE are 1, 2, and 3. A small number of records had the value 0, which is not defined. These were mapped to category 3 (Others).

2.2.5 Remapping the SEX Column

The original coding was 1 for Male and 2 for Female. This was remapped to 0 for Male and 1 for Female, which is a standard binary encoding that is easier for machine learning algorithms to work with.

2.2.6 One-Hot Encoding

EDUCATION and MARRIAGE are nominal categorical variables. Their numeric codes do not imply any ranking or order. Using them as raw numbers could mislead a model into thinking a higher education code means something more. To avoid this, we applied one-hot encoding, which creates a separate binary column for each category.

2.2.7 Handling Infinity Values

During feature engineering, some division operations produced very large or infinite values due to edge cases, for example when bill amounts were zero or negative. These infinity values were replaced with zero before model training to ensure all algorithms could run without errors.

Chapter 3. Descriptive and Statistical Analysis

3.1 Basic Summary Statistics

We started by computing standard descriptive statistics for all numeric features. The table below summarises the key columns:

Feature	Mean	Std Dev	Min	Max
LIMIT_BAL	167,484	129,748	10,000	1,000,000
AGE	35.49	9.22	21	79
BILL_AMT1	51,223	73,636	−165,580	964,511
PAY_AMT1	5,664	16,563	0	873,552

The credit limit ranges widely from NT$10,000 to NT$1,000,000. Most customers are in their 30s. Bill amounts can be negative, which typically represents credit adjustments or refunds issued by the bank.

3.2 Default Rate Across Groups

We looked at how the default rate changes across different customer groups.

- **By Gender:** Male customers defaulted at a rate of 24.2%, while female customers defaulted at 20.8%. There is a small but noticeable gap.
- **By Education:** High school educated customers had the highest default rate at 25.2%, followed by university students at 23.7%, and graduate school holders at 19.2%. The "Others" category had an unusually low rate of 7.1%, but this group is small and mixed.
- **By Credit Limit:** Customers who did not default had an average credit limit of around NT$178,000, while defaulters averaged around NT$130,000. This makes intuitive sense since banks typically assign higher limits to more reliable customers.

3.3 Correlation with Default

Looking at pairwise correlations between features and the target variable DEFAULT, the repayment status columns (PAY_0, PAY_2, etc.) showed the strongest positive correlation. This means customers who delay payments more often are more likely to default. LIMIT_BAL showed a negative correlation, meaning higher credit limits are associated with fewer defaults.

Chapter 4. Visualisations and Inferences

4.1 Target Class Distribution

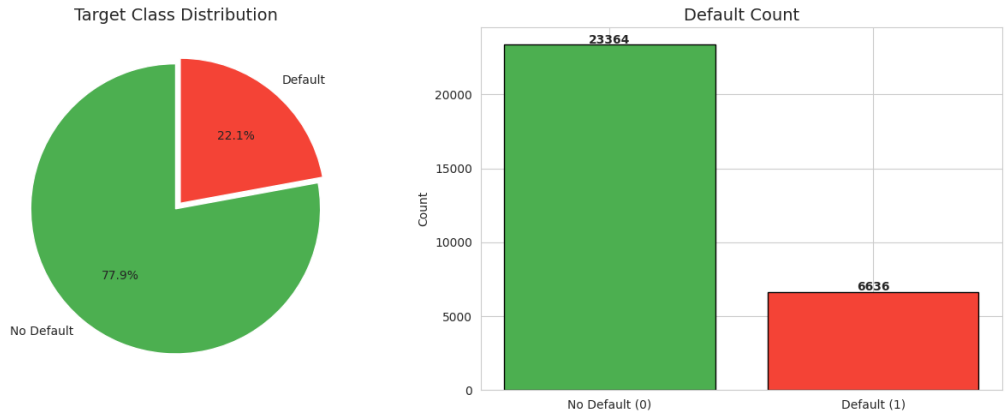

Figure 4.1: Target class distribution showing 77.9% non-defaults and 22.1% defaults.

Out of 30,000 customers, 23,364 did not default and 6,636 defaulted. The dataset is imbalanced at 22.1% default rate. This matters because a model that always predicts "no default" would already score 77.9% accuracy without learning anything useful. So we need to look at other metrics like F1-score and ROC-AUC.

4.2 Default Rate by Education and Gender

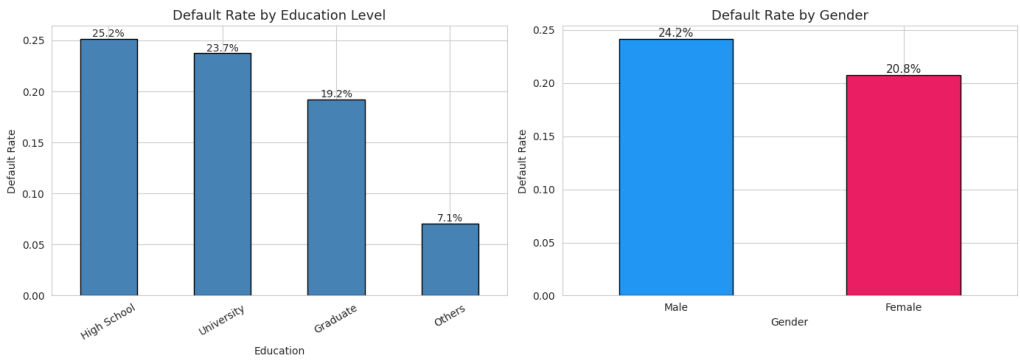

Figure 4.2: Default rate by education level (left) and by gender (right).

High school educated customers default most frequently at 25.2%, while graduate school holders default the least among the main groups at 19.2%. Males default slightly more than females. These differences exist but are not large enough for education or gender to be reliable standalone predictors.

4.3 Age Distribution

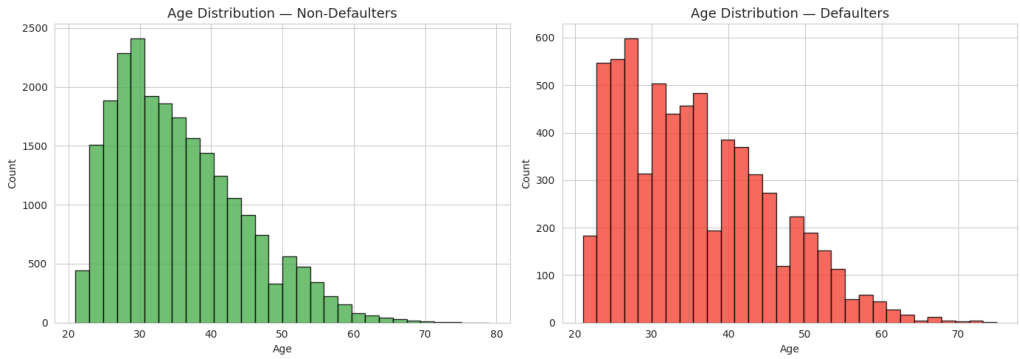

Figure 4.3: Age distribution for non-defaulters (left, green) and defaulters (right, red).

Both defaulters and non-defaulters follow a similar age distribution, peaking around 25 to 35 years old. The shapes are nearly identical, which tells us age is not a strong differentiating feature on its own.

4.4 Credit Limit by Default Status

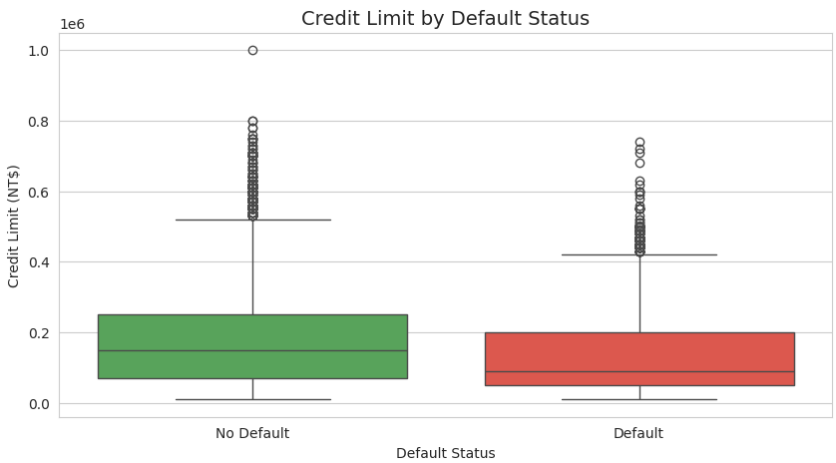

Figure 4.4: Boxplot comparing credit limits between defaulters and non-defaulters.

Non-defaulting customers have a visibly higher median credit limit compared to defaulting customers. There is also a wider spread of high-limit outliers in the non-default group. This confirms that credit limit is a useful signal: customers with lower limits are at higher risk.

4.5 Repayment Status Analysis

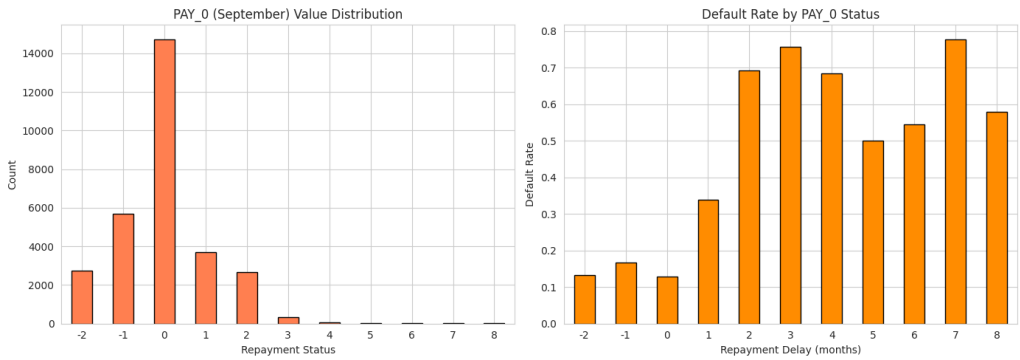

Figure 4.5: PAY_0 value counts (left) and default rate by PAY_0 status (right).

This is one of the most informative plots in the entire analysis. The right chart makes it very clear: as payment delay increases, the default rate rises sharply. Customers with a 2-month delay defaulted at close to 70%. Those with a 3-month delay defaulted at over 75%. Customers who paid on time (-1 or -2) had very low default rates around 13 to 16%. PAY_0 is clearly one of the strongest individual predictors.

4.6 Correlation Heatmap

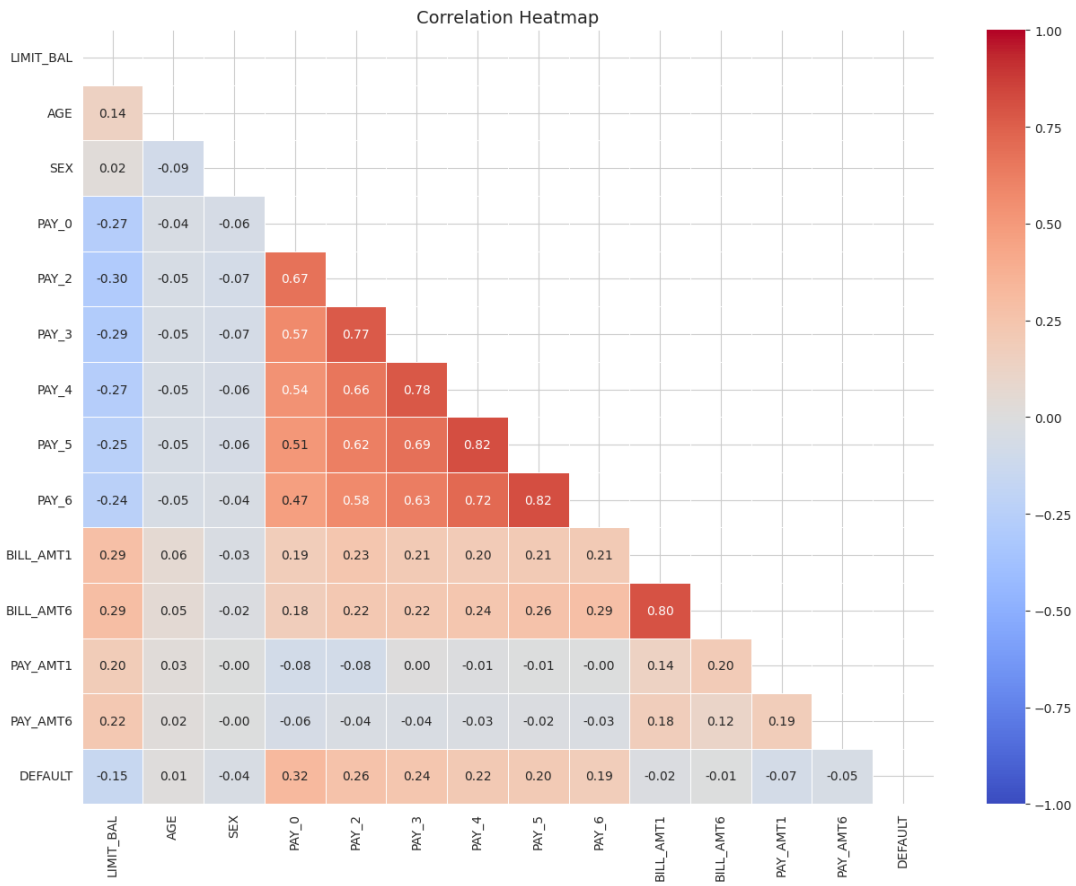

Figure 4.6: Correlation heatmap of key features.

The PAY columns are all positively correlated with DEFAULT. Consecutive PAY columns are also highly correlated with each other (for example, PAY_2 and PAY_3 have a correlation of 0.77), which is expected since customers who delayed one month often delay the next. LIMIT_BAL is negatively correlated with DEFAULT at -0.15. Bill amounts are strongly correlated with each other since they represent a running balance.

4.7 Bill Amount and Payment Trends Over 6 Months

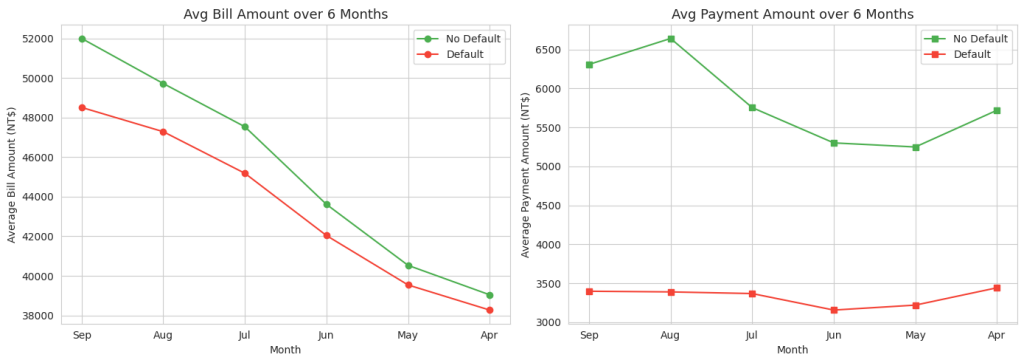

Figure 4.7: Average bill amount (left) and average payment amount (right) over 6 months.

Non-defaulters actually have slightly higher average bill amounts, which seems counter-intuitive. But looking at the right chart, non-defaulters also pay significantly more each month (around NT$5,500 to 6,500). Defaulters pay very little each month (around NT$3,200 to 3,500) and their payments are flat and unchanging. This means defaulters are not keeping up with their bills, and the unpaid balance grows month by month.

Chapter 5. Hypothesis and Research Questions

5.1 Hypotheses Framed

Before running machine learning models, we listed specific hypotheses based on our exploratory analysis and understanding of credit default behaviour. These helped us stay focused and gave us a way to validate our findings at the end.

- **H1:** Customers with higher repayment delays (PAY_0 > 0) are significantly more likely to default.
- **H2:** Customers with lower credit limits are more likely to default.
- **H3:** Customers who consistently pay less than their bill amount are at higher default risk.
- **H4:** Younger customers (age 25–35) have a higher default rate than older customers.
- **H5:** Education level has a meaningful effect on the probability of default.
- **H6:** A low bill-to-payment ratio (low repayment efficiency) is a strong predictor of default.

5.2 Research Questions

1. Which demographic group (gender, age, education) carries the highest default risk?
2. How strongly do repayment patterns (PAY_0 to PAY_6) predict default?
3. Can we build a model with accuracy above 80% to predict credit defaults?
4. Which features are the most important predictors of default?
5. Is there a credit limit threshold below which default risk sharply increases?

Chapter 6. Feature Engineering

6.1 Why We Created New Features

The original dataset captures repayment behaviour across 6 months through 18 separate columns (6 PAY, 6 BILL_AMT, 6 PAY_AMT). While these contain useful information, using all of them individually adds noise. We aggregated them into 9 summary features that capture the essential patterns more cleanly and meaningfully.

6.2 Engineered Features

Feature	Description
AVG_PAY_DELAY	Average repayment delay across 6 months. Higher values mean more consistent late payments.
DELAY_COUNT	Number of months (out of 6) where the customer was late on payment.
MAX_PAY_DELAY	The worst single-month delay the customer had across 6 months.
AVG_BILL_AMT	Average bill statement amount over 6 months.
AVG_PAY_AMT	Average monthly payment made by the customer.
AVG_PAY_RATIO	Average fraction of the bill that was actually paid. Captures repayment efficiency.
TOTAL_OUTSTANDING	Total unpaid balance across all 6 months (bill minus payment, summed).
DEFICIENCY_COUNT	Number of months where the customer paid less than their bill amount.
UTILIZATION	Average ratio of bill amount to credit limit, measuring how much credit is being used.

After creating these features, the original 18 raw columns were dropped since their information is now captured more compactly.

6.3 Feature Distributions

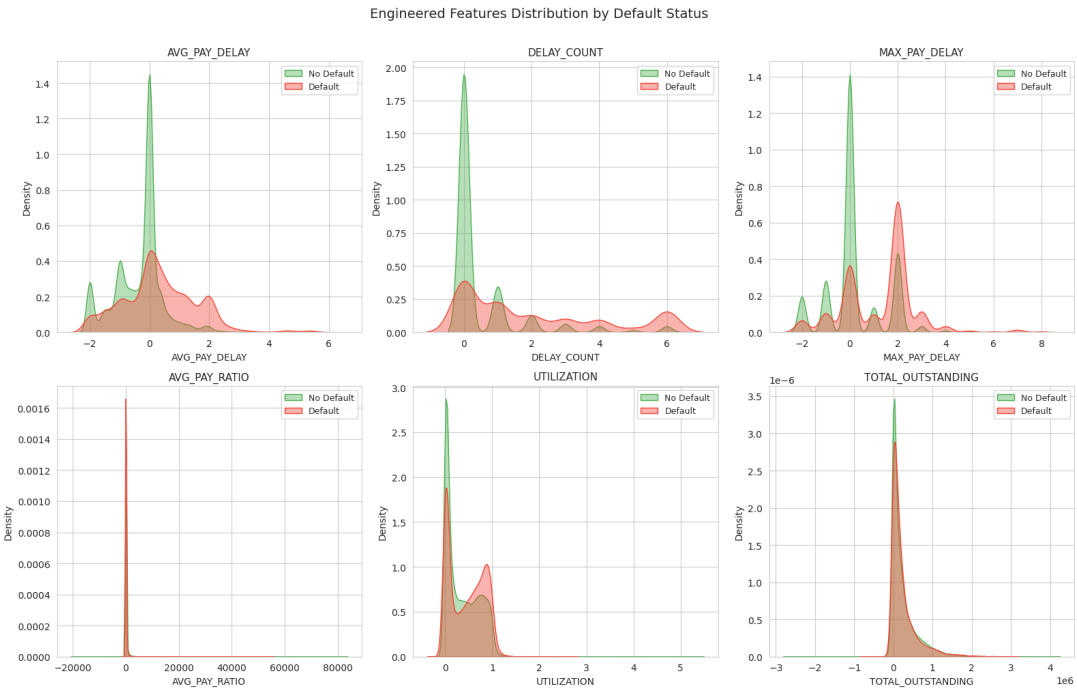

Figure 6.1: Distribution of engineered features for defaulters vs non-defaulters.

The density plots show that AVG_PAY_DELAY and DELAY_COUNT provide the clearest separation between defaulters and non-defaulters. Non-defaulters cluster tightly near zero for both features, while defaulters are spread out towards higher values. UTILIZATION also shows some separation. AVG_PAY_RATIO shows a large spike near zero for defaulters, meaning many of them paid almost nothing relative to their bill.

Chapter 7. Machine Learning Classification

7.1 Algorithm Selection and Justification

We trained and compared four classification algorithms. Using multiple algorithms lets us compare performance and understand which approach is best suited for this problem.

Algorithm	Why We Used It
Logistic Regression	A simple and interpretable baseline. Works well for binary classification and produces probability outputs directly.
Decision Tree	Can capture non-linear relationships in the data. Easy to understand. Used as a single-model baseline for tree-based methods.
Random Forest	An ensemble of many decision trees. More robust and less prone to overfitting than a single tree. Handles noisy and mixed features well.
XGBoost	A gradient boosting method that builds trees sequentially, where each tree corrects the mistakes of the previous one. Well-known for strong performance on tabular data.

XGBoost was selected as the primary model because it consistently outperforms simpler methods on structured tabular data and deals reasonably well with class imbalance.

7.2 How the Models Were Trained

The data was split into 80% for training and 20% for testing using stratified sampling. Stratified splitting ensures that the 22.1% default rate is maintained in both the training and test sets. For Logistic Regression, features were scaled using StandardScaler since it is sensitive to the magnitude of input values. The tree-based models (Decision Tree, Random Forest, XGBoost) do not require scaling.

Chapter 8. Model Evaluation and Results

8.1 Performance Comparison

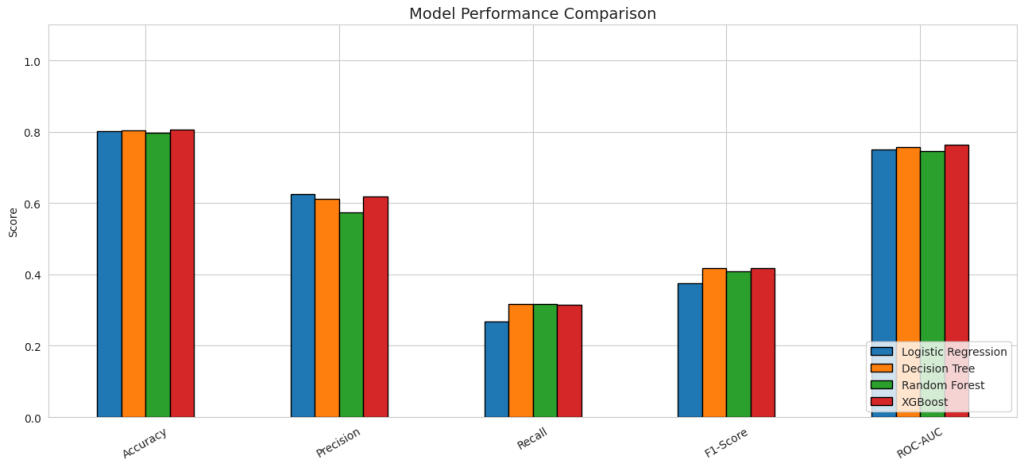

Figure 8.1: Comparison of all four models across five evaluation metrics.

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.8013	0.6331	0.2688	0.3775	0.7513
Decision Tree	0.8053	0.6128	0.3170	0.4180	0.7562
Random Forest	0.8043	0.5743	0.3181	0.4090	0.7450
XGBoost	**0.8082**	**0.6181**	**0.3143**	**0.4163**	**0.7640**

All four models achieved roughly 80% accuracy. XGBoost had the best ROC-AUC at 0.764 and the best overall balance across metrics. However, the recall values are relatively low for all models, meaning a fair number of actual defaulters are not caught. This is expected with imbalanced data where the model sees many more non-defaults during training.

8.2 ROC Curves

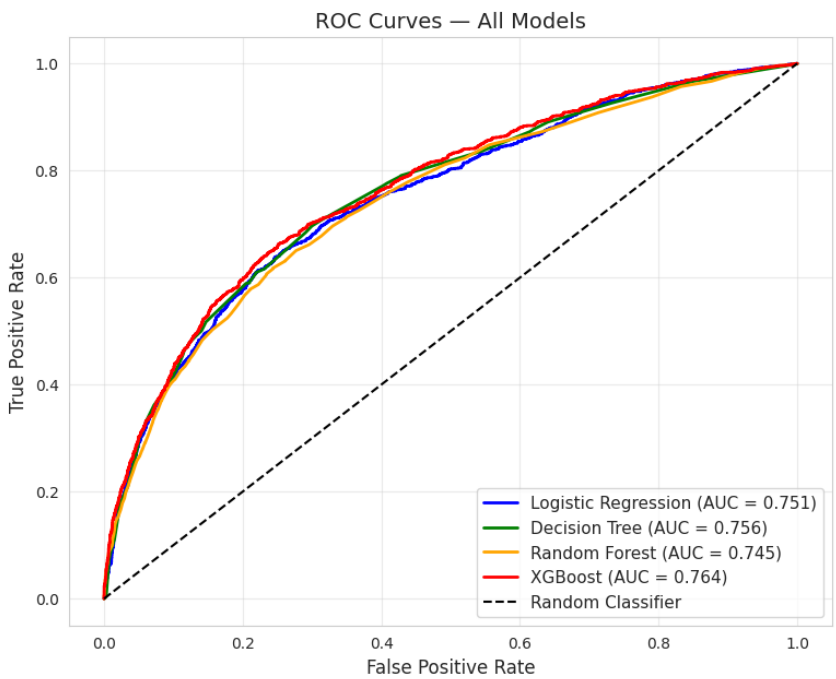

Figure 8.2: ROC curves for all four models on the test set.

The ROC curve plots the true positive rate against the false positive rate at different decision thresholds. A curve closer to the top-left corner is better. XGBoost (shown in red) slightly leads the other models with an AUC of 0.764. All models perform well above the diagonal random baseline.

8.3 Confusion Matrices

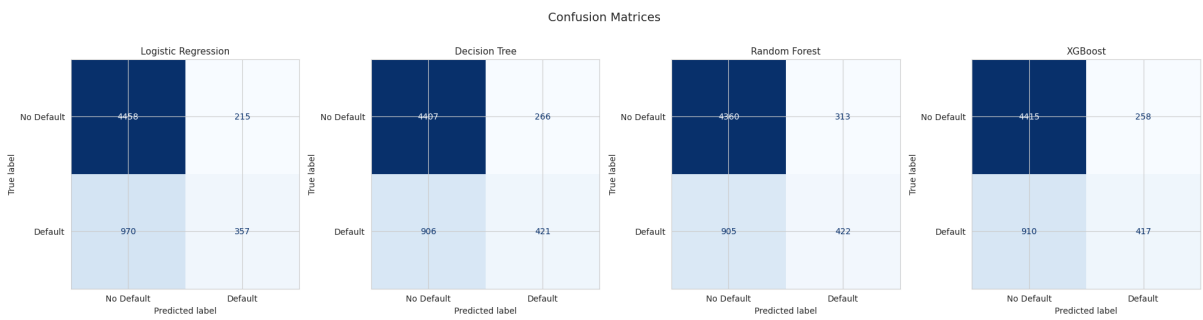

Figure 8.3: Confusion matrices for all four models on the test set.

All models correctly identify most non-defaulters. However, all models also miss a significant number of actual defaulters. For example, XGBoost correctly identifies 417 defaulters but misses 910. This is the key challenge with class imbalance: since non-defaults

dominate the training data, models are biased towards predicting that class. Improving recall would require techniques like oversampling the minority class or adjusting class weights.

8.4 XGBoost Classification Report

Class	Precision	Recall	F1-Score	Support
No Default	0.83	0.94	0.88	4673
Default	0.62	0.31	0.42	1327
Accuracy				0.81
Macro avg	0.73	0.63	0.65	6000
Weighted avg	0.79	0.81	0.79	6000

The model performs much better on the non-default class (F1 = 0.88) compared to the default class (F1 = 0.42). The precision of 0.62 for the default class means that when the model does predict a default, it is correct about 62% of the time.

8.5 Cross-Validation

To verify that the model results are not due to a lucky train-test split, we performed 5-fold stratified cross-validation on XGBoost. The model achieved a mean ROC-AUC of approximately 0.77 with low variance across folds. This confirms that the performance is stable and generalises well to unseen data.

Chapter 9. Feature Importance

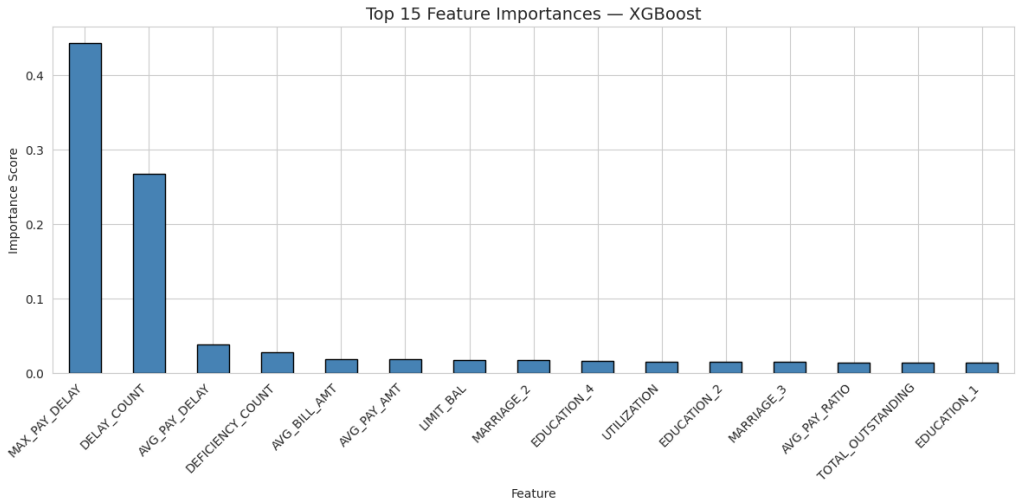

Figure 9.1: Top 15 feature importances as reported by XGBoost.

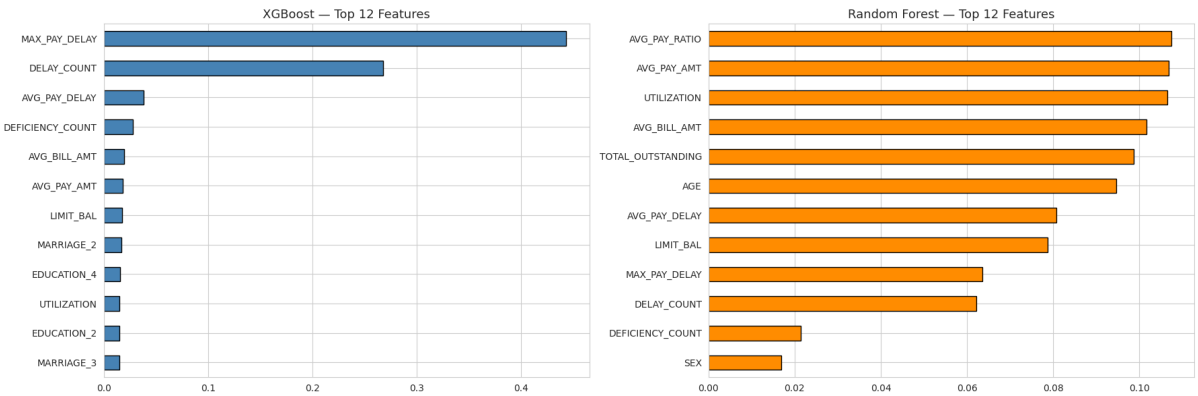

Figure 9.2: Feature importance comparison between XGBoost (left) and Random Forest (right).

9.1 Key Observations

XGBoost assigns extremely high importance to MAX_PAY_DELAY (0.44) and DELAY_COUNT (0.27). These two features alone account for over 70% of the model's decision-making. This means XGBoost is essentially asking: how bad was the worst delay, and how many months in total did the customer delay? If both answers are high,

the model flags the customer as a default risk. The remaining features contribute smaller but still useful amounts.

Random Forest distributes importance much more evenly across features. AVG_PAY_RATIO, AVG_PAY_AMT, UTILIZATION, AVG_BILL_AMT, and TOTAL_OUTSTANDING all score similarly. This means Random Forest considers a broader set of financial behaviours rather than focusing on one or two signals.

Both models consistently rank payment delay features (MAX_PAY_DELAY, DELAY_COUNT, AVG_PAY_DELAY) among the most important, which gives strong confidence that these are the genuinely predictive signals in this dataset.

Chapter 10. Conclusions

10.1 Summary of Findings

This project built and evaluated a machine learning system to predict credit card default using 6 months of billing and repayment history from 30,000 customers.

From the data analysis:

- The dataset is moderately imbalanced with a 22.1% default rate.
- Repayment delay (especially PAY_0) is the single strongest predictor. Customers who were 2 months late defaulted at nearly 70%.
- Non-defaulters carry higher credit limits, consistent with the idea that banks assign limits based on creditworthiness.
- Defaulters pay much less relative to their bills each month, causing debt to grow over time.
- Education level and gender have some effect, but they are not strong standalone predictors.

From machine learning:

- All four models achieved around 80% accuracy.
- XGBoost performed best with a ROC-AUC of 0.764 and the highest F1-score.
- Low recall is a common challenge across all models due to class imbalance.
- The engineered features MAX_PAY_DELAY and DELAY_COUNT were the most informative features by a wide margin.

10.2 Hypotheses Revisited

ID	Hypothesis	Outcome
H1	Repayment delay predicts default	Strongly confirmed
H2	Lower credit limit increases default risk	Confirmed
H3	Paying less than bill amount increases risk	Confirmed
H4	Younger customers default more	Partially supported; age alone is a weak predictor
H5	Education level affects default probability	Partially supported; effect is present but small
H6	Low repayment efficiency predicts default	Confirmed

10.3 Research Questions Answered

1. **Highest risk group:** High school educated male customers showed the highest default rates. However, demographic features on their own are weak predictors. Behavioural patterns (payment delays) matter far more.
2. **Repayment pattern importance:** PAY columns and derived features like MAX_PAY_DELAY dominate the model. They are the most predictive signals by a large margin.
3. **Above 80% accuracy:** Yes, XGBoost achieved 80.8%. However, given the class imbalance, ROC-AUC of 0.764 is a more meaningful measure of how well the model actually discriminates.
4. **Most important features:** MAX_PAY_DELAY, DELAY_COUNT, and AVG_PAY_DELAY are the top three.
5. **Credit limit threshold:** While no exact threshold was computed, the analysis suggests that customers with limits below NT$100,000 are at noticeably higher risk.

10.4 Business Recommendation

Based on this analysis, credit providers can build a practical early warning system. Customers who show the following signs should be flagged for early intervention:

- Two or more consecutive months of payment delay

- Monthly payment amounts that are consistently much lower than bill amounts
- High credit utilization over multiple months
- A rising total outstanding balance month over month

Early actions such as sending payment reminders, offering restructured payment plans, or placing temporary credit holds could help reduce defaults and protect both the customer and the bank from financial loss.

Appendix: Code Summary

The complete code for this project is written in Python 3 and organised as a Jupyter Notebook with 13 clearly labelled sections. The main libraries used are listed below:

- **pandas, numpy** — data loading, manipulation, and feature engineering
- **matplotlib, seaborn** — all visualisations and plots
- **scikit-learn** — train/test split, Logistic Regression, Decision Tree, Random Forest, StandardScaler, and all evaluation metrics
- **xgboost** — XGBoost classifier

The notebook follows the same structure as this report. Each section contains comments explaining what the code does and why. All plots were saved as PNG files and are included in this document.